

SASWAT K. NAYAK

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EXPERIENCE

Data Scientist(Full Time) Innovaccer Analytics(Digital Tech Unicorn)

July 2020-July 2022

- Optimized the healthcare database system by migrating the knowledge base from PostgreSQL to Neo4j, resulting in an 80% reduction in query time, enhancing the efficiency of data retrieval and improving stakeholder satisfaction.
- Collaborated with a team of 3 interns to develop an NLP pipeline for extracting structured data from EHRs, leveraging NLTK and SpaCy to reduce data extraction time by 90% and operational costs by 50%.
- Built interactive dashboards using Streamlit and Bayesian Inference to analyze hospital performance metrics and healthcare trends, empowering stakeholders with actionable insights.
- Built and deployed a robust containerized task queue subsystem using Docker and Kubernetes, enabling automatic scaling across dozens of EC2 instances and improving task processing efficiency by 60%.
- Followed agile methodologies (Scrum, Kanban) to plan and prioritize tasks, ensuring timely completion of projects.
- Reviewed and refactored code across multiple projects, maintaining 95% unit test coverage and ensuring adherence to software engineering best practices.

Projects [Streamlit Visualization](#)

- Execute an end-to-end data engineering [project](#) on real-time stock market data using Kafka, Amazon Web Services (AWS), Apache Kafka, Glue, Athena, and SQL.

Developed a personalized [chatbot](#) using LangChain and Llama2, achieving a 70% improvement in response relevance by integrating RAG (Retrieval-Augmented Generation), vector stores, and optimized document chunking.

- Designed and implemented a high-performance file management system with relational database integration, improving CRUD operation efficiency by 50% and providing a user-friendly web interface for seamless data management.
- Fine-tuned **Mistral-7B** on **GPT-4** generated rationales, enabling reasoning in Mistral-7B by **knowledge distillation**. Fine-tuned Mistral-7B on the call classification dataset by using **AWQ quantization, LoRA** and **SFT**.
- Built an emergency SOS Voice Agent with LlamaIndex Agent RAG and Bolna Voice Agent.
- Scraped Nordstrom using Selenium for a GenAI project, rotating proxies to prevent website from scrape-blocking and multi-threading for improved scraping speeds- [Github](#).

RELEVANT COURSES

Software Engineering, Image Processing, Advanced Machine Learning, Statistics, Information Retrieval, Databases, Probability and Stochastic Processes, Regression and Time Series Analysis, Machine Learning in Materials, Algorithms and Data Structures, Linear Algebra, Graph Databases, Generative AI.

SKILLS

Javascript, Python, CI/CD, Docker, Gitlab,R, C++, SQL, Tableau, Clinical Trials, Spark, GPU, Alignment-handbook, MergeKit, Angular, React, Java, Agile, Redis, GCL, Eval-harness, Azure, Caboodle, A/B testing, Excel, Power BI, Networking, Operating Systems, Computer Graphics, Node.js, RDkit Data, AWS Batch, NLP, GANs, RNN, Gen AI, Tensorflow, PyTorch, GANs, Airflow, Statistics, LLM, Pandas, Numpy, Sagemaker, Large Language Models(BERT, GPT-2, T5, RoBERTa), Google TTS, GBM, XGBoost, Kubernetes, Flink, Langchain, Transformers, MLops, HuggingFace.

Research Assistant: GenAI in 3D Photonics, Semiconductor Design September 2022 - Dec 2024 — *Madison, WI*

- Introduced novel spatial descriptors for 3D lattices that captured more information than the traditional symmetry based descriptors and outperformed them in machine learning tasks by 50%(F1 Score). Used kernel PCA to find the latent space where we can find a hyperplane to linearly separate classes.

Machine Learning Intern, Ushur

Nov 2020 - Dec 2020

- Designed and implemented an advanced algorithm combining Google Tesseract OCR with rule-based techniques to accurately extract customer data from insurance claim documents; securing victory in an internal hackathon.